

1. Python is easy to learn and code because:

- a) It has a simple syntax that is close to English
- b) It has a complex syntax that requires deep knowledge
- c) It uses only numbers and symbols
- d) It requires memorizing complex commands

Ans: (a)

2. Python is considered a high-level language because:

- a) It is close to machine code
- b) It requires the user to manage memory manually
- c) It abstracts away complex details of the computer's hardware
- d) It directly interacts with the operating system

Ans: (a)

3. Which of the following statements is true about Python's cost?

- a) Python is a paid language with a subscription
- b) Python is free to use, but requires a license for distribution
- c) Python is open-source and free to use

Ans: (c)

4. Python is platform-independent, which means:

- a) Python code can only run on Windows
- b) Python code must be rewritten for each platform
- c) Python code can run on any platform that has a compatible Python interpreter
- d) Python code can only run on Linux

Ans: (c)

5. Python is considered an interpreted language because:

- a) Python code is compiled into machine code before execution
- b) Python code is converted into bytecode and executed by a virtual machine at runtime
- c) Python code is executed line by line by the system hardware

d) Python code is pre-executed in the system memory

Ans: (c)

6. Python is a dynamically typed language, which means:

- a) Variables must be explicitly declared with their type before use
- b) The type of a variable is determined at runtime and can change during execution
- c) Variables must always remain the same type once declared
- d) Types are checked at compile time, not runtime

Ans: (c)

7. Write about single value datatype and its type with example

Ans: single value datatype is datatype that store only one single value in the variable.

Types :

1) Numeric –

- a. integer- eg. a=100, b=200
- b. float – a=1.02, b=4.5
- c. complex- 20-4j

2) Boolean – a=True, b=False

8. How do you define a string in Python?

Ans: it is the collection of characters (numbers, alphabets, and symbols) enclosed in the single, double and triple quotes.

9. Explain the use of double quote and triple quote with example.

Ans: double quotes is used when already the single quotes is present in the string.

eg. String="India has world's largest population"

Triple quotes is used when we want to continue string in the next line

eg. String= """ India has world's largest population,
all over the universe"""

10. How can you access individual characters in a string?

Ans: string[ind]

11. Explain the concept of string indexing in Python.

Ans: Indexing is the sub address given to the each and every block of the string

2 Types: Right to left (0, 1, 2, 3, 4,)

Left to right (.....-1, -2, -3)

12. Which expression is used to convert the negative index value to positive index?

Ans: $+ve_index = len(string) + (-ve_index)$

13. What is the difference between indexing and index()?

Ans: Indexing: It is help to fetch the character at the given position.

Index(): It is help to fetch the index of the given character

14. Explain the upper() and lower() string methods.

Ans: upper(): It convert the all character of string to the uppercase(capital).

lower(): It convert the all character of string to the lowercase (small).

15. What is the purpose of the strip() method?

Ans: Strip() is used to remove the specific characters from the string,

Lstrip() – lstrip() remove character from left of the string.

rstrip() – rstrip() remove character from right of the string.

16. How can you check if a string starts or ends with a specific substring?

Ans :

17. What does the replace() method do in Python strings?

Ans: replace() is used to modify the string inplace of any characters from the string.

syntax: `variable.replace("old_string", "new_string")`

18. Describe the usage of the split() method.

Ans: Split() break down the string into multiple string when the space (or " ", "_",etc) occurs.

19. Explain the use of the format() method for string formatting with example.

Ans: format () is used to print string in the customised way and pass arguments.

`name ='Mansi'`

age =27

Print("My name is { } and I am { } year old". Format("name, age"))

20.What is f-string formatting? Provide an example.

Ans: f-string is print string in the customised way by passing variable into the string.

Print("My name is { name } and I am { year } year old")

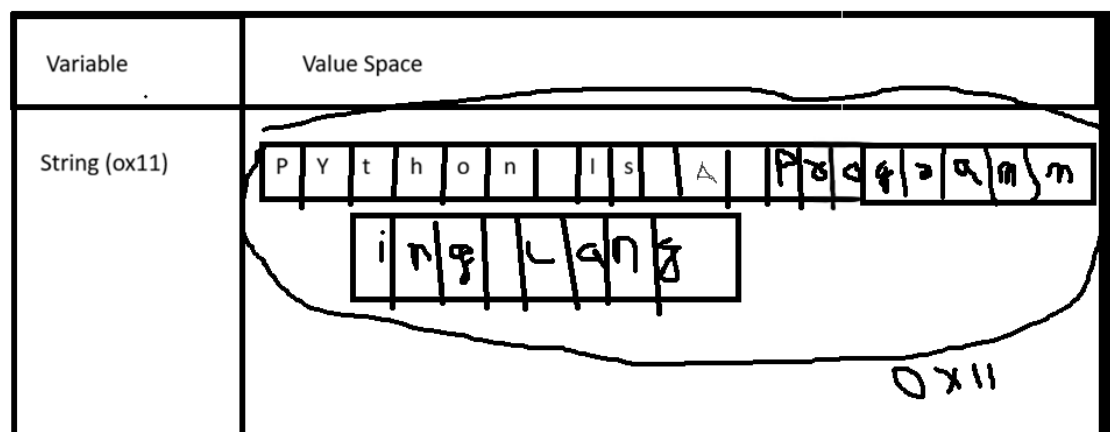
21. What is difference between index() and find().

Ans : index() helps to fetch the index position of the given character.

Find() helps to fetch the lowest index position of the given character.

22.Explain the memory allocation of string using diagram

string ='python is a programming lang'



23. Explain multiple variable creation with example.

Ans: Var1, var2, = val1, val2, a, b= 10, 20

24. Explain rules for creating variable with example.

Ans:

1. The keyword cannot be used as a variable
True=90
2. Variable should not start with number
2A=44
3. Variable should not begin with the space or have space in between
A b=akshata
4. Variable should not contain special character

@a=aniket

5. The var Can Any alphabets In Upper Case Lowercase or both or the alphanumeric
Ans=123
6. According to industrial standard rule the length of the variable is restricted to 32
String=bdjbsbdb.....32

25. Explain the memory allocation of list using diagram

```
_list = ['hello', ['python', 'SeleniUm', 89, [34, 'web', ['good eveNing', 'good', 78], 90.8]], 'sql', 89-9j]
```

26. get the following output

1. ['good eveNing', 'good', 78]

```
_list [1][3][2]
```

2. 'good'

```
_list [1][3][2][2]
```

3. 'U'

```
_list[1][1][6]
```

4. 'N'

```
_list [1][3][2][0][9]
```

5. 90.8

```
_list [1][4]
```

7. 78

```
_list [1][3][2][0][2]
```

27. Explain the difference between append(), extend() and insert() with example

Ans: append() – adds the elements at the end of the list.

```
Eg. list = [1, 2, 3, 4, 5, 6]
list.append(2026)
print(list) → [1, 2, 3, 4, 5, 6, 2026]
```

extend() – adds the MVD as a individual char into the list.

```
Eg. list = [1, 2, 3, 4, 5, 6]
list.append("Mona")
print(list) → [1, 2, 3, 4, 5, 6, 2026, 'm', 'o', 'n', 'a']
```

insert() – adds the element at the specific index position.

```
Eg. List = [1, 2, 3, 4, 5, 6]
list.append(2, 'Happy')
```

```
print(list) → [1, 2, "Happy ", 3, 4, 5, 6, 2026]
```

28. Explain the difference between pop() and remove() with example

Ans: Pop method removes an item at a specific position. If you do not give it a number, it deletes the last item.

```
fruits = ["apple", "banana", "cherry"]  
fruits.pop(1)  
print(fruits) → ["apple", "cherry"]
```

The remove() method searches for a specific text or number and deletes the first match it finds.

```
fruits = ["apple", "banana", "cherry"]  
fruits.remove(banana)  
print(fruits) → ["apple", "cherry"]
```

29. difference between add and update with example

30. difference between symmetric_difference_update and symmetric_difference with example

31. difference between remove and discard with example

Ans: remove will erase the given character only if present in the set, if not present it will get error

```
Eg, List = {1, 2, 3, 4, 5, 6}  
list.remove(2)  
print(list) → {1, 3, 4, 5, 6}
```

discard also erase the given character if, present, if not present it ignores

```
Eg, List = {1, 2, 3, 4, 5, 6}  
list.discard(2026)  
print(list) → {1, 2, 3, 4, 5, 6}
```

32. mention the rules of set type

Ans:

- We cannot insert mutable data items inside the set
- The set are unordered in nature so we cannot use indexing on the set type
- Set Mutable in nature
- The default value of dictionary is { }
- Default value of set is Boolean false

33. mention the rules of keys of dict

Ans:

- The keys of dictionary should always be immutable in nature

- The values can be either mutable or immutable
- The keys of dictionary should always be unique in nature
- The default value of dictionary is { }
- Dictionary Is internally Boolean false
- Key always contains new overwritten values to the key
- We can use key to fetch the value
- This helps to add a new key value pair and also modify the existing key.

34. difference between pop() and popitem() with example

Ans: Pop removes the element from the beginning of the Dictionary

Eg, dict = {1: 10, 2: 30, 'Name': "Akshata"}
 print(pop()) → {2: 30, 'Name': "Akshata"}

pop item remove the specific element Or the character Tune into

Eg, dict = {1: 10, 2: 30, 'Name': "Akshata"}
 print(popitem ('Name')) → {1: 10, 2: 30}

35. Given this nested list:

```
L = [3, 7, [1, 4, 'hello']]
# Reassign "hello" to be "goodbye"
L[2][2]= "goodbye"
```

36. Using keys and indexing, grab the 'hello' from the following dictionaries:

```
d1 = {'simple_key': 'hello'}
print(d1['simple_key'])

d2 = {'k1': {'k2': 'hello'}}
d2.get('hello')

d3 = {'k1': [{'nest_key': ['this is deep', ['hello']]]}]}
```

37. You are given two variables:

```
age = 4
name = "Sammy"
```

38. Use print formatting to print the following string:[both .format and fstring]

```
"Hello my dog's name is Sammy and he is 4 years old"
Print( f " Hello my dog's {name} is Sammy and he is {year} years old")
```

```
Print( "Hello my dog's { } is Sammy and he is { } years old".format(name, age))
```

39. Write a program to replace one string with another. e.g. "Hello World"

```
# replaces "World" with "Universe"
```

```
Print(string.replace("world", "Universe"))
```

40. Write a program to swap two numbers without using the 3rd variable.

```
a = 10
```

```
b = 20
```

```
a, b = b, a
```

```
print(a, b)
```

```
##a = 20
```

```
##b = 10
```